

Cathleen A. Goodwin

California Regional Water Quality Control Board
North Coast Region

ORDER NO. 89-117
I.D. NO. 1B88049RSON

WASTE DISCHARGE REQUIREMENTS

FOR

UNIVERSITY OF CALIFORNIA BODEGA MARINE LABORATORY
DOMESTIC WASTEWATER TREATMENT AND DISPOSAL SYSTEM

Sonoma County

The California Regional Water Quality Control Board, North Coast Region (hereinafter the Board) finds that:

1. The University of California, Bodega Marine Laboratory (hereinafter the discharger) submitted a Report of Waste Discharge dated May 6, 1988 for proposed domestic waste collection, treatment, and disposal facilities designed to accommodate wastes from the discharger's research and administration facilities, and housing facilities. Additional information to complete the Report of Waste Discharge was submitted on August 8, 1989. The Report of Waste Discharge includes design reports for the discharger's proposed wastewater facilities dated April, 1988, July, 1989 and August, 1989. The facilities are located west of the community of Bodega Bay on Bodega Head, in T6N, R11W, MDB&M, as shown on Attachment "A" incorporated herein and made a part of this Order. The discharger's property is bordered by the Pacific Ocean and the nearshore waters of the Bodega Marine Life Refuge (a designated Area of Special Biological Significance) to the west and Bodega Bay to the east.
2. The discharger proposes to construct two septic tank-leachfield systems to handle domestic wastewater generated by the research and administration facilities and housing facilities. The proposed system for the research and administration facilities includes two septic tanks, a lift station and a conventional leachfield trench system designed to handle an average daily flow of 12,120 gallons per day (gpd) and a peak flow of 20,000 gpd. The proposed system for the housing facilities includes a combined gravity/pressure collection system, a central lift station, and four subsurface absorption beds. The proposed housing system is designed to handle an average daily flow of 6000 gpd and a peak flow of 9600 gpd. Both proposed subsurface systems include a pressure distribution system for uniform dosing.
3. The proposed domestic wastewater collection, treatment, and disposal facilities have been designed to handle the domestic wastewater flows that will be generated at the completion of the discharger's long-range facility expansion. The proposed domestic wastewater collection, treatment, and disposal facilities will replace the existing septic tank-leachfield systems which currently serve the

administration, research and housing facilities. The existing facilities are located in areas of seasonally high groundwater and are not adequate to meet the increasing domestic wastewater flows.

4. The wastewater discharge from the discharger's flow-through aquarium system is regulated by NPDES Permit/Waste Discharge Requirements Order No. 89-11. Small quantities of chemicals used in the research laboratories are transported for off-site disposal in accordance with state and federal regulations.
5. The Board adopted Water Quality Control Plans for the Klamath River Basin (1A) and the North Coastal Basin (1B) on March 20, 1975. The Klamath River Basin Plan (1A) was combined with the North Coastal Basin Plan (1B) to form the Water Quality Control Plan for the North Coast Region. The Plan for the North Coast Region was adopted by the Board on April 28, 1988, and approved by the State Water Resources Control Board on November 15, 1988. The Plan includes water quality objectives and receiving water limitations.
6. The beneficial uses of the Pacific Ocean include:
 - a. navigation
 - b. water contact recreation
 - c. noncontact water recreation
 - d. ocean commercial and sport fishing
 - e. preservation of areas of special biological significance
 - f. wildlife habitat
 - g. preservation of rare and endangered species
 - h. marine habitat
 - i. fish migration
 - j. fish spawning
 - k. shellfish harvesting
7. The beneficial uses of the waters of the Bodega Marine Life Refuge area include:
 - a. scientific research and education
 - b. esthetic enjoyment
 - c. preservation and enhancement of fish, wildlife, and other aquatic resources or preserves
8. The beneficial uses of Bodega Bay include:
 - a. agricultural water supply
 - b. industrial service water supply
 - c. navigation
 - d. water contact recreation
 - e. noncontact water recreation
 - f. ocean commercial and sport fishing
 - g. cold freshwater habitat
 - h. preservation of areas of special scientific significance
 - i. wildlife habitat

- j. marine habitat
- k. fish migration
- l. fish spawning
- m. shellfish harvesting

9. The beneficial uses of areal groundwater include:

- a. municipal and domestic water supply
- b. industrial water supply

10. The Regents of the University of California certified a Final Environmental Impact report (FEIR) for the long-range development of the University of California Bodega Marine Laboratory in compliance with the California Environmental Quality Act. The FEIR provides an evaluation of the environmental effects of domestic wastewater disposal at the Bodega Marine Laboratory facility. The Regional Board finds that the implementation of all mitigation measures described in the FEIR and this Order, will mitigate any potential adverse water quality impacts that may result from the proposed project.
11. The Board has notified the discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations.
12. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

THEREFORE, IT IS HEREBY ORDERED that the the University of California, Bodega Marine Laboratory shall comply with the following:

A. DISCHARGE PROHIBITIONS

- 1. The discharge of waste to land that is not under the control of the discharger, except as authorized under "C. Solids Disposal", is prohibited.
- 2. The discharge of waste to surface waters is prohibited.
- 3. The discharge of all wastewater shall be kept underground at all times.
- 4. The discharge of any pollutant not identified and authorized by this Order, or the discharge of any pollutant in a manner or quantity which differs from that stated in the report of waste discharge is prohibited.
- 5. An adequate replacement area equivalent to the initial disposal area shall be reserved at the time of the initial installation of all leachfields. Incompatible uses of the initial disposal area and the replacement area are prohibited.
- 6. Creation of a pollution, contamination, or nuisance, as defined by Section 13050 of the California Water Code (CWC), is prohibited [Health and Safety Code, Section 5411]

B. DISCHARGE SPECIFICATIONS

1. The average daily dry-weather flow of wastewater to the research and administration facilities system shall not exceed 12,120 gpd averaged over a period of 30 consecutive days.
2. The peak wastewater flow to the research and administration facilities system shall not exceed 20,000 gpd.
3. The average daily dry-weather flow of wastewater to the housing system shall not exceed 6000 gpd averaged over a period of 30 consecutive days.
4. The peak wastewater flow to the housing system shall not exceed 9600 gpd.

C. SOLIDS DISPOSAL

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of at a legal point of disposal, and in accordance with the provisions of Title 23, Chapter 3, Subchapter 15 of the Code of Regulations or as waived pursuant to Section 13269 of the California Water Code.

D. PROVISIONS

1. A copy of this Order shall be maintained at the discharge facility and be available at all times to operating personnel.
2. Severability

Provisions of these waste discharge requirements are severable. If any provision of these requirements is found invalid, the remainder of these requirements shall not be affected.

3. Operation and Maintenance

The discharger must maintain in good working order and operate as efficiently as possible any facility or control system installed by the discharger to achieve compliance with the waste discharge requirements.

4. Revisions of Requirements

The Board will review this Order periodically and may revise requirements when necessary.

5. Vested Rights

This Order does not convey any property rights of any sort, nor any exclusive privileges. The requirements prescribed herein do not authorize the commission of any act causing inquiry to persons or property, nor protect the discharger from his liability under federal, state, or local laws, nor create a vested right for the discharger to continue the waste discharge.

6. Inspections

The discharger shall permit authorized staff of the Board:

- a. entry upon premises in which an effluent source is located or in which any required records are kept;
- b. access to copy any records required to be kept under terms and conditions of this Order;
- c. inspection of monitoring equipment or records; and
- d. sampling of any discharge.

7. Noncompliance

In the event the discharger is unable to comply with any of the conditions of this Order due to:

- a. breakdown of waste treatment equipment;
- b. accidents caused by human error or negligence; or
- c. other causes such as acts of nature;

The discharger must notify the Executive Officer by telephone as soon as he or his agents have knowledge of the incident and confirm this notification in writing within two weeks of the telephone notification. The written notification shall include pertinent information explaining reasons for the noncompliance and shall indicate what steps are being taken to prevent the problem from recurring.

8. Change in Discharge

In the event of a material change in the character, location, or volume of a discharge, the discharger shall file with this Board a new report of waste discharge at least 120 days before making any such change.

9. Monitoring

The discharger must comply with the Contingency Planning and Notification Requirements Order No. 74-151 and the Monitoring and Reporting Program No. 89-117 and any modifications to these documents as specified by the Executive Officer. Such documents are attached to this Order and incorporated herein.

Chemical, Bacteriological, and bioassay analyses shall be conducted at a laboratory certified for such analysis by the State Department of Health Services. In the event a certified laboratory is not available to the discharger, analysis performed by a noncertified laboratory will be accepted provided:

- a. A quality assurance/quality control program is instituted by the laboratory. A manual containing the steps followed in this program must be kept in the laboratory and available for inspection by staff of the Board, the quality assurance/quality control program must conform to EPA guidelines or procedures approved by the Board.

- b. The laboratory will become certified within the shortest practicable time if the State certification program is resumed.

10. Change in Ownership

In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the discharger, the discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this Board.

Certification

I, Benjamin D. Kor, Executive Officer,
do hereby certify that the foregoing is
~~a full, true, and correct copy of an~~
Order adopted by the California Regional
Water Quality Control Board, North Coast
Region, on November 16, 1989.

ORIGINAL SIGNED BY

Benjamin D. Kor
Executive Officer

(bmldmwdr)

California Regional Water Quality Control Board
North Coast Region

MONITORING AND REPORTING PROGRAM NO. 89-117

FOR

UNIVERSITY OF CALIFORNIA BODEGA MARINE LABORATORY
DOMESTIC WASTEWATER TREATMENT AND DISPOSAL SYSTEM

Sonoma County

MONITORING

1. Waste Volume

The discharger shall measure and record the monthly volume of wastewater discharged to the two main pumping stations.

2. Septic Tank and Pump System Inspection

- a. The discharger shall inspect the septic tanks annually to determine the need for pumping. When the level of solids in any tank is equal to or in excess of one third of the total depth of the tank, the tank shall be pumped.
- b. The discharger shall inspect all wet wells and pumps monthly and shall record the maintenance provided to them.

3. Disposal Field Inspections and Monitoring

- a. The discharger shall inspect the disposal fields weekly and shall note and record any odors, evidence of surfacing wastewater or other signs of malfunction.
- b. The discharger shall record the level of wastewater in the inspection wells located in the leachfield and absorption beds monthly.

4. Groundwater Monitoring

- a. The discharger shall measure and record the elevation of water in all monitoring wells on a monthly basis. Reference datum shall be ground surface at the well.
- b. The discharger shall sample standing water in monitoring wells MW-A, MW-B, MW-C and MW-D (Attachment "B") and monitoring wells H-3 and H-10 (Attachment "C") for analysis of the following:

<u>Constituent</u>	<u>Units</u>	<u>Sampling Frequency</u>
Total Coliform	MPN/100 ml	Quarterly ^{1/}
Fecal Coliform	MPN/100 ml	Quarterly
Nitrate Nitrogen	mg/l	Quarterly

REPORTING

Annual monitoring reports shall be submitted by the fifteenth of August each year. Quarterly monitoring reports shall be submitted by the fifteenth of February, May, August, and November of each year. Monthly monitoring reports shall be submitted to the Regional Board for each month by the 15th day of the following month. The report shall include all items specified above and the dates on which monitoring or inspection was performed.

The discharger shall have all flow measuring devices tested annually and their accuracy certified. This certification shall be submitted annually with the monitoring report for the month of August.

In reporting the monitoring data, the discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements. During periods of no discharge, the reports shall certify no discharge.

1/ Quarterly samples shall be collected in January, April, July and October of each year.

Ordered by ORIGINAL SIGNED BY

Benjamin D. Kor
Executive Officer

November 16, 1989

California Regional Water Quality Control Board
North Coast Region

CONTINGENCY PLANNING AND NOTIFICATION REQUIREMENTS

FOR

ACCIDENTAL SPILLS AND DISCHARGES

ORDER NO. 74-151

The California Regional Water Quality Control Board, North Coast Region, finds that:

1. Section 13225 of the Porter-Cologne Water Quality Control Act requires the Regional Board to perform general duties to assure positive water quality control.
2. The Regional Board has been advised of situations in which preparations for, and response to accidental discharges and spills have been inadequate.
3. Persons discharging waste or conveying, supplying, storing, or managing wastes or hazardous materials have the primary responsibility for contingency planning, incident reporting and continuous and diligent action to abate the effects of such unintentional or accidental discharge.

THEREFORE, IT IS HEREBY ORDERED THAT:

- I. All persons who discharge wastes or convey, supply, store, or otherwise manage wastes or other hazardous material shall:
 - A. Prepare and submit to this Regional Board, according to a time schedule prescribed by the Executive Officer, a contingency plan defining the following:
 1. Potential locations and/or circumstances under which accidental discharge incidents might be expected to occur,
 2. Possible water quality effects of accidental discharges,
 3. The conceptual plan for cleanup and abatement of accidental discharge incidents, including:
 - a. The individual who will be in charge of cleanup and abatement activities on behalf of the discharger,
 - b. The equipment and manpower available to the discharger to implement the cleanup and abatement plans,
 - B. Immediately report to the Regional Board any accidental discharge incidents. Such notification shall be made by telephone as soon as the responsible person or his agent has knowledge of the incident.

- C. Immediately begin diligent and continuous action to cleanup and abate the effects of any unintentional or accidental discharge. Such action shall include temporary measures to abate the discharge prior to completing permanent repairs to damaged facilities.
 - D. Confirm the telephone notification in writing within two weeks of the telephone notification. The written notification shall include: reasons for the discharge, duration and volume of the discharge, steps taken to correct the problem and steps being taken to prevent the problem from recurring.
- II. Upon original receipt of phone report (I.B), the Executive Officer shall immediately notify all affected agencies and known users of waters affected by the unintentional or accidental discharge.
- III. Provide updated information to the Regional Board in the event of change of staff, size of the facility, or change of operating procedures which will affect the previously established contingency plan.
- IV. The Executive Officer or his employees shall maintain liaison with the discharger and other affected agencies and persons to provide assistance in cleanup and abatement activities.
- V. The Executive Officer shall transmit copies of this Order to all persons whose discharges of waste handling activities are governed by Waste Discharge Requirements or an NPDES Permit. Such transmittal shall include a current listing of telephone numbers of the Executive Officer and his key employees to facilitate compliance with Item I.B of this Order.

Ordered by


Benjamin D. Kor
Executive Officer

July 24, 1974
(Retyped January, 1986)

Your primary notification should be to the Regional Board office at Santa Rosa at (707) 576-2220. During off hours, you will be able to leave a recorded message at that number and, if you have a spill or discharge emergency, you will also be referred to the State Office of Emergency Services (OES) at (800) 852-7550. OES maintains a roster of key employees and will relay your notification to Regional Board staff.

California Regional Water Quality Control Board
North Coast Region

GENERAL MONITORING AND REPORTING PROVISIONS

February 28, 1973
(Revised on April 19, 1983)

GENERAL PROVISIONS FOR SAMPLING AND ANALYSIS

Unless otherwise noted, all sampling and sample preservation, shall be in accordance with the current edition of "Standard Methods for the Examination of Water and Wastewater". All analyses shall be conducted in accordance with the test procedures approved under 40 CFR Part 136 "Test Procedures for the Analysis of Pollutants".

All analyses shall be performed in a laboratory certified to perform such analyses by the California State Department of Health or a laboratory approved by the Executive Officer.

All samples shall be representative of the waste discharge. Effluent samples shall be taken downstream of any addition to the treatment works and prior to mixing with the receiving waters.

The discharger shall calibrate and perform maintenance procedures on all monitoring instruments and equipment to ensure accuracy of measurements.

GENERAL PROVISIONS FOR REPORTING

For every item where the requirements are not met, the discharger shall submit a statement of the actions undertaken or proposed which will bring the discharge into full compliance with requirements at the earliest time and submit a timetable for correction.

By January 30 of each year, the discharger shall submit an annual report to the Regional Board. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

The discharger shall maintain all sampling and analytical results, including strip charts; date, exact place, and time sampling; date analyses were performed; analyst name, analytical techniques used; and result of all analyses. Such records shall be retained for a minimum of three years. This period of retention shall be extended during the course of any unresolved litigation regarding this discharge.

Monitoring reports shall be signed by:

- a. In the case of corporations, by a principal executive officer at least of the level of vice-president or his duly authorized representative, if such representative is responsible for the overall operation of the facility from which the discharge originates.

General Monitoring and
Reporting Provisions

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- b. In the case of a partnership, by a general partner.
- c. In the case of a sole proprietorship, by the proprietor.
- d. In the case of a municipal, state or other public facility, by either a principal executive officer, ranking elected official, or other authorized employee.